VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS101

Programming for Problem Solving & Object-Oriented Principles - Comprehen

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Syllabus Focus: CS101 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Programming for Problem Solving & Object-Orie

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Imperative Foundations & Memory Layout

" Syllabus Scope: Data representations, bitwise operators, execution models, control flow, functions, cal

Unit 2: Pointers & Dynamic Memory Management

" Syllabus Scope: Pointer arithmetic, arrays, heap allocation malloc/free/new/delete, memory leak prevent

CS101: Programming for Problem Solving & Object-Oriented Principles - Advanced Topic

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Object-Oriented Architecture

" Encapsulation, abstraction, constructor/destructor chains, operator overloading, deep vs shallow copying

Unit 4: Inheritance & Dynamic Polymorphism

" Single and multiple inheritance, virtual function tables vtable, abstract base classes, runtime type identif

Unit 5: Templates, Generic Programming & STL

" Function and class templates, Standard Template Library vector, map, set, custom iterators, exception handli

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Illustrate virtual table vtable and vptr mechanics during dynamic method dispatch in C++ with memory d
2. Differentiate stack vs heap allocation, explaining memory fragmentation and RAII principles.
3. Formulate a generic template class for a thread-safe circular buffer with boundary condition handling.

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Bjarne Stroustrup - The C++ Programming Language Addison-Wesley 4th Ed
* Robert Lafore - Object-Oriented Programming in C++ Sams Publishing

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